

ROBUSTNESS OF MORPHING ATTACK DETECTION TO COMPRESSION

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ROBUSTNESS OF S-MAD TO COMPRESSION

Morphing attacks

Accomplice



Morph



Attacker



1) Accomplice succeeds in having the morphed image included in a genuine ID document

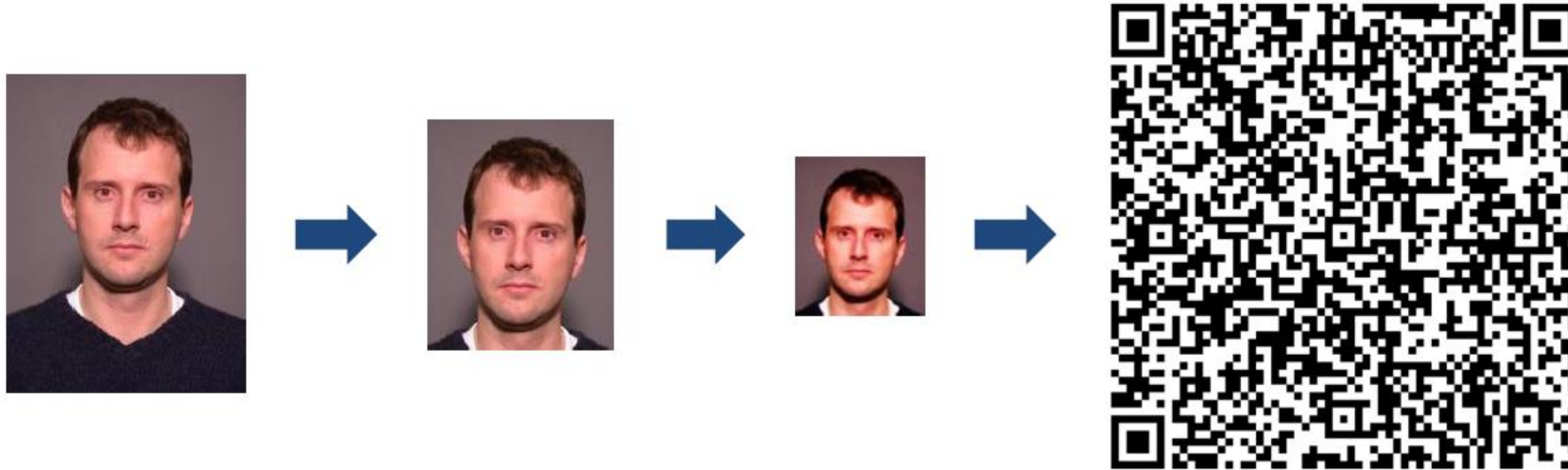


2) Attacker uses morphed ID-doc. to gain unauthorised entry



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An extreme use case



- At 1100 bytes, we may still be able to recognise faces
- But can we still detect morphs?

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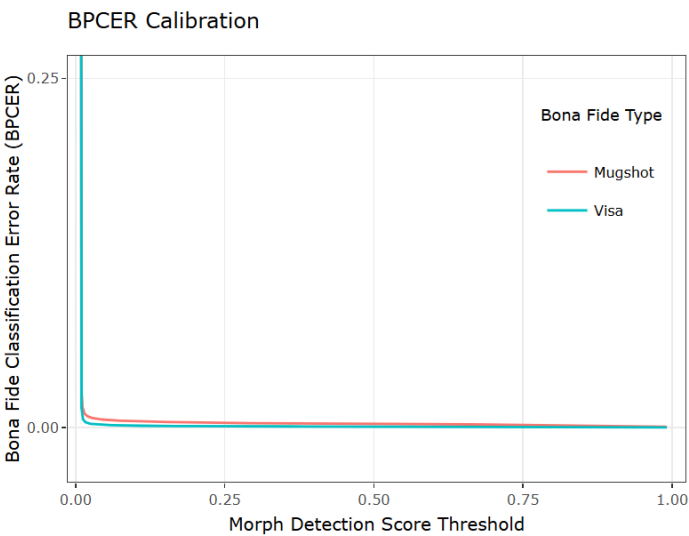
The study: algorithm + operational threshold

Algorithm (+reportcard)	Submission Date	Type	Manual [Tier 3 - High Quality]	Print+Scanned [Tier 3 - High Quality]	Local Morph Colorized Match [Tier 2 - Automated]	Visa-Border [Tier 2 - Automated]	Lincoln [Tier 3 - High Quality]	DST [Tier 3 - High Quality]
idemia 004	2024-12-16	single- image	0.526	0.041	0.000	0.001	0.620	0.626
idemia 002	2024-03-20	differential	0.743	0.455	0.454	0.148	-	-
idemia 003	2024-05-18	differential	0.743	0.179	0.314	0.111	-	-
idemia 001	2024-01-25	differential	0.759	0.440	0.815	0.585	-	-
secunet 003	2025-02-21	differential	0.848	0.433	0.805	0.847	-	-

MACER@BPCER_m=0.003

MACER: Morphing Attack Classification Error Rate (i.e. False Negatives)
(i.e. probability that an attacker will fool the detection algorithm)

BPCER: *Bona fide* Presentation Classification Error Rate (i.e. False Positives)
(i.e. probability that a genuine document is rejected due to false morph-detection)



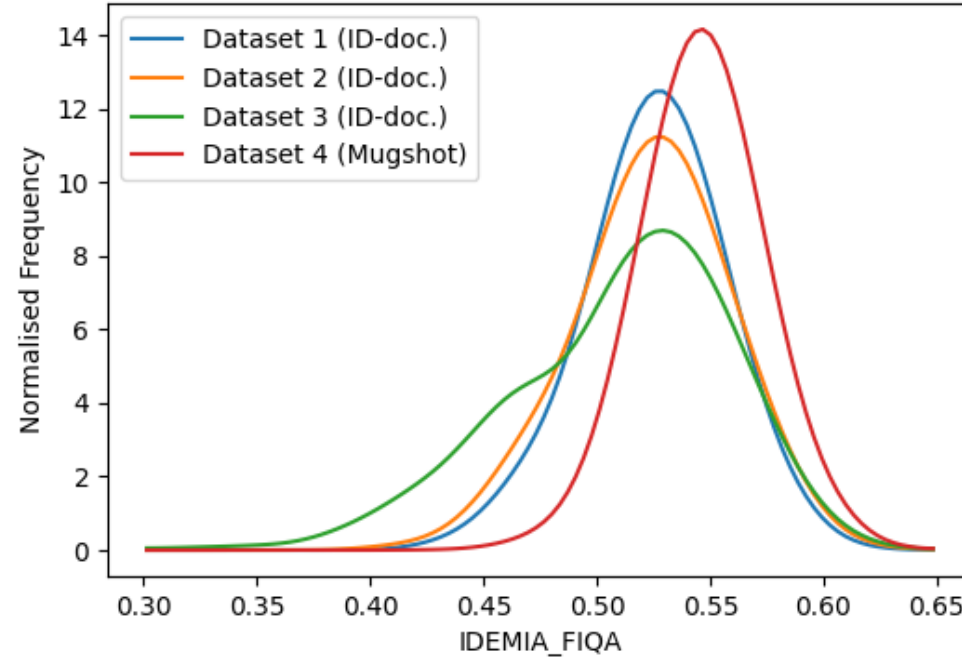
BPCER_m=0.01 → ~0.015

We will use this value as the “operational threshold” throughout this study

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The study: **datasets + metrics**

Dataset	Image-type	Num. images
Dataset 1	ID-doc.	4619
Dataset 2	ID-doc.	5000
Dataset 3	ID-doc.	5000
Dataset 4	Mugshot	8069



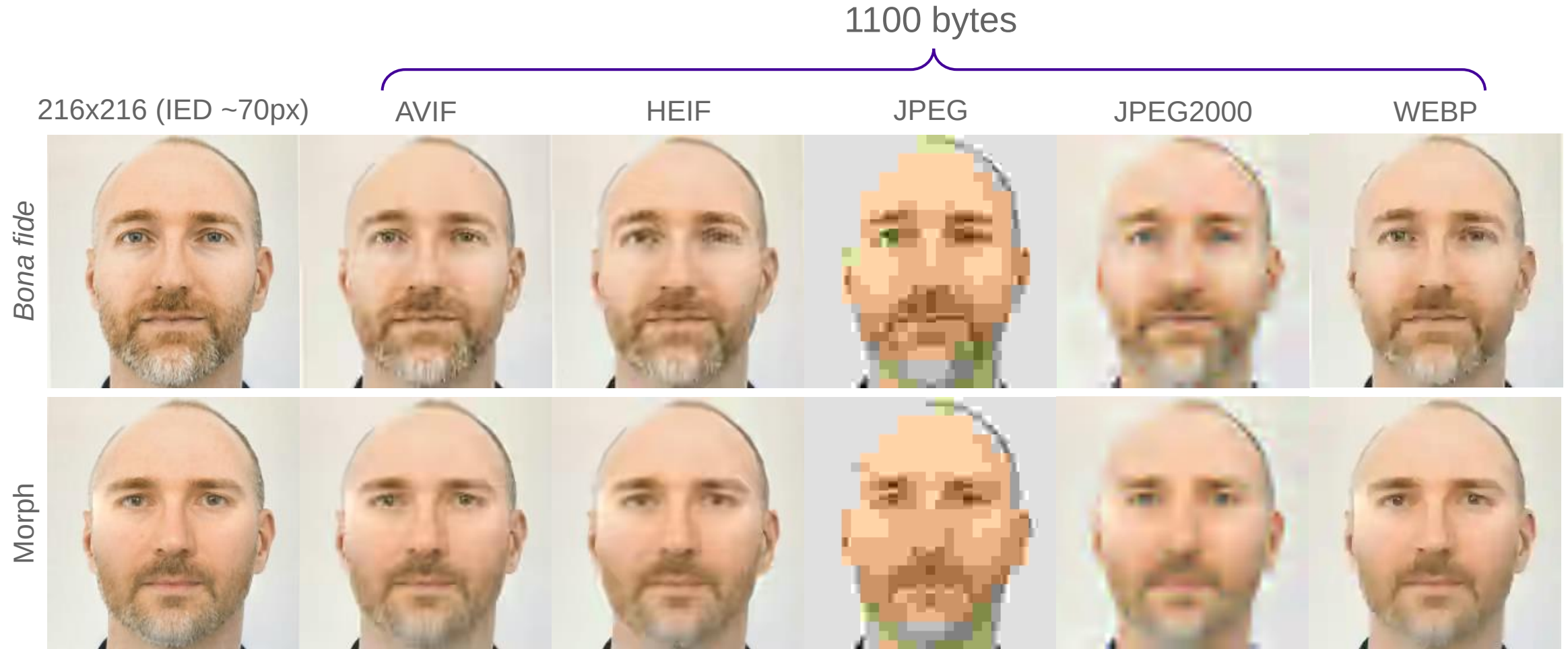
Metrics:

For each of these datasets we will look at the effect of compression on values of...

- MACER@BPCER_{m=0.01}
- BPCER@BPCER_{m=0.01}

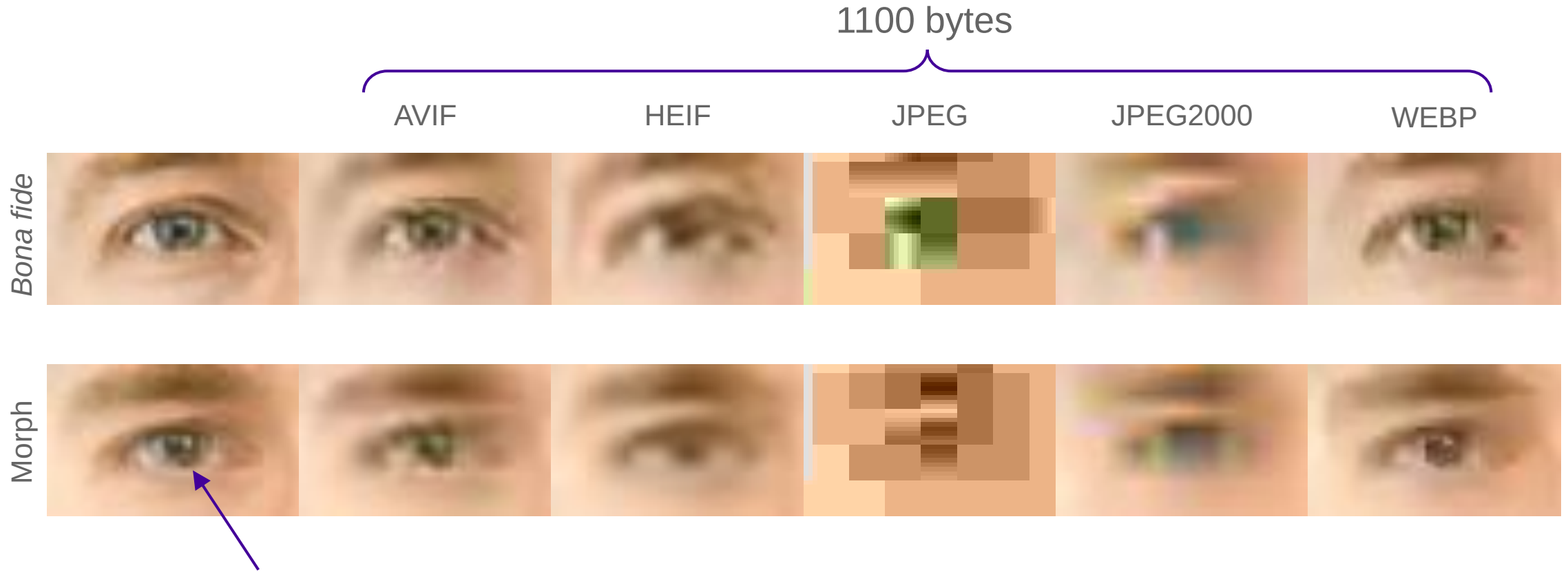
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The study: crop + compression types



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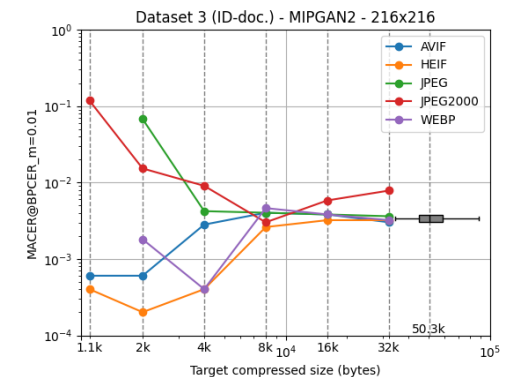
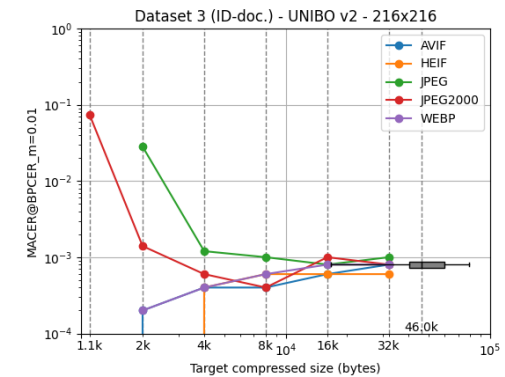
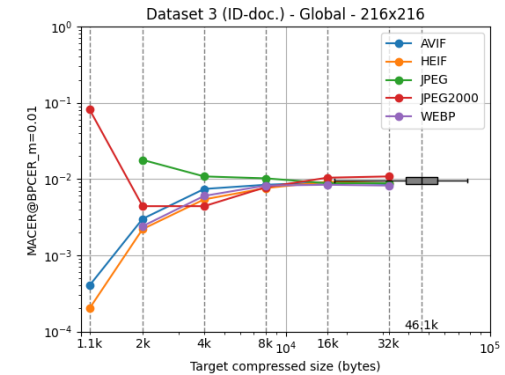
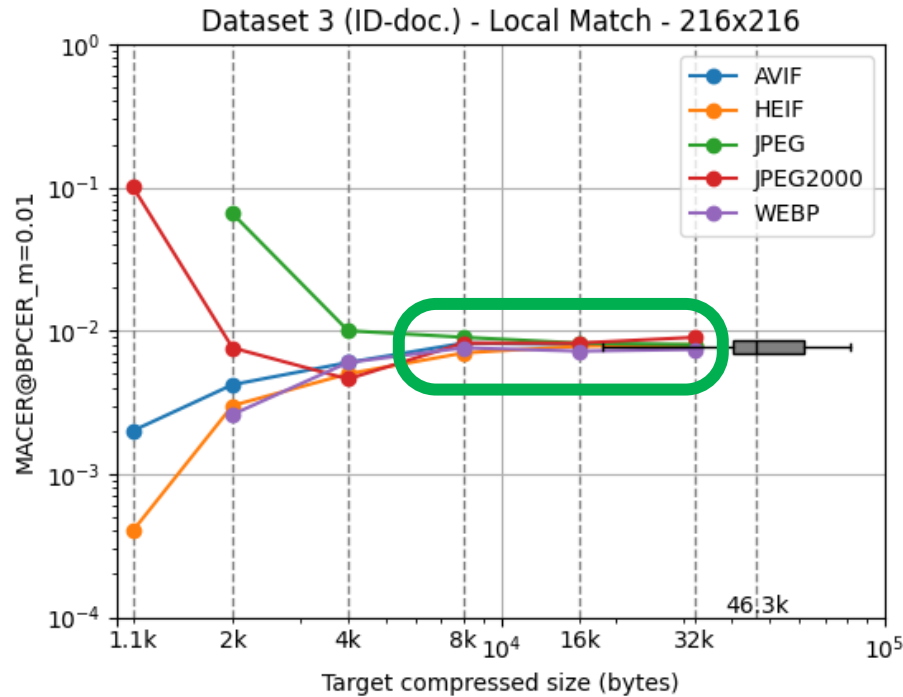
Compression types - zoomed



Artifact visible in
uncompressed morphed
image

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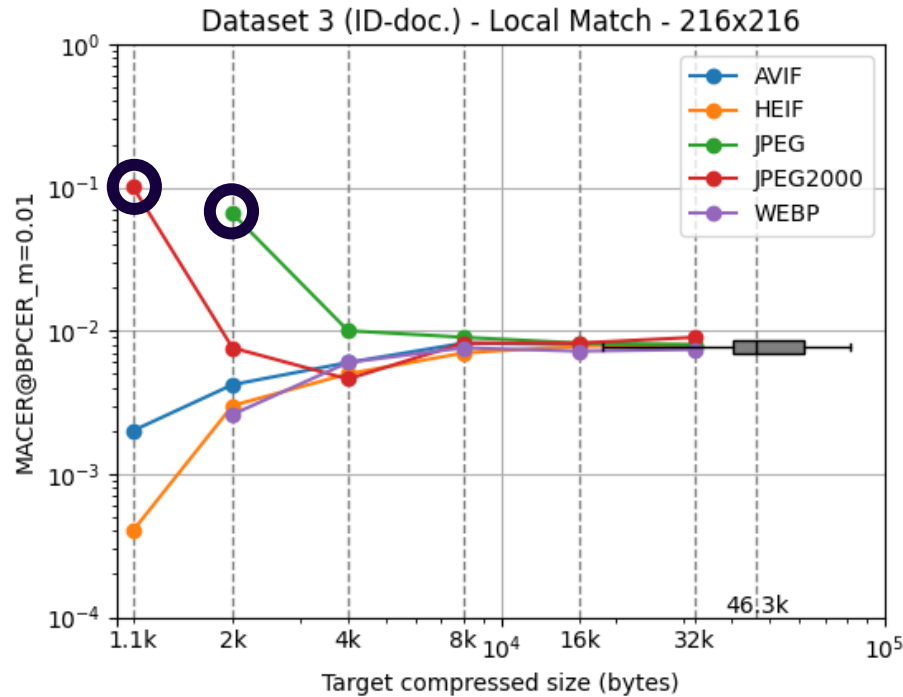
Does compression obscure morphing artifacts?



- Light compression does not increase MACER
 - an attacker can't rely on compression to mask artifacts
- Heavy compression can increase or decrease the rate of detections, depending on the compression type.

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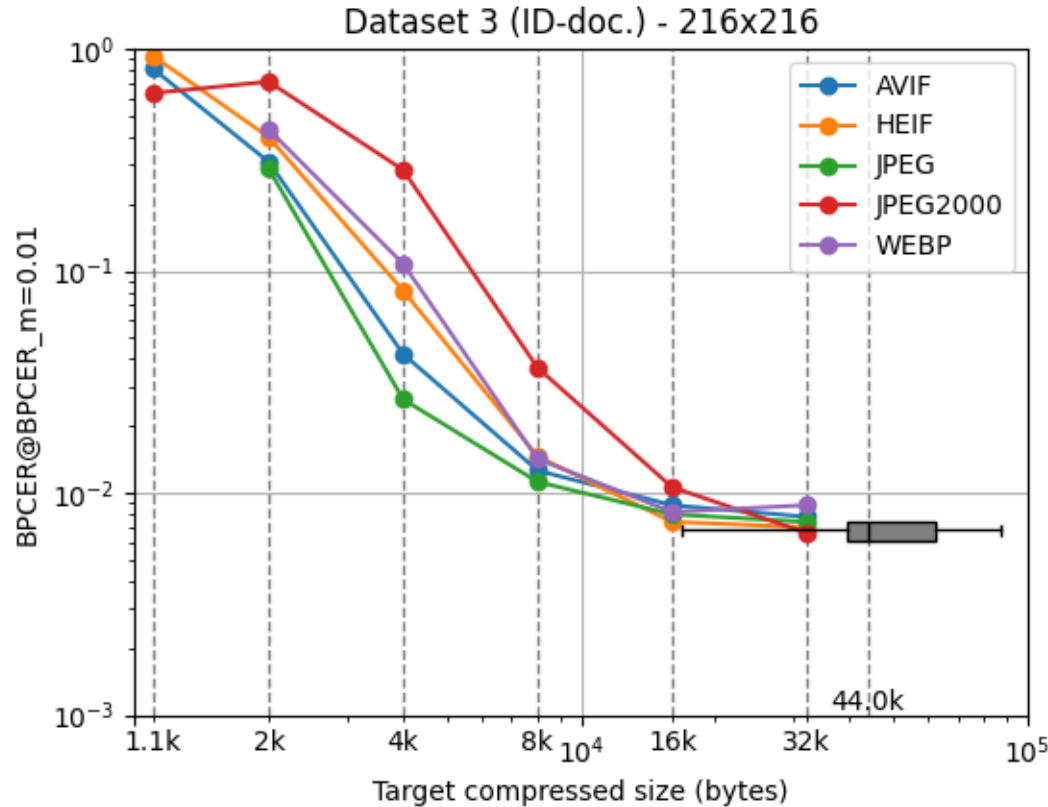
Does compression obscure morphing artifacts?



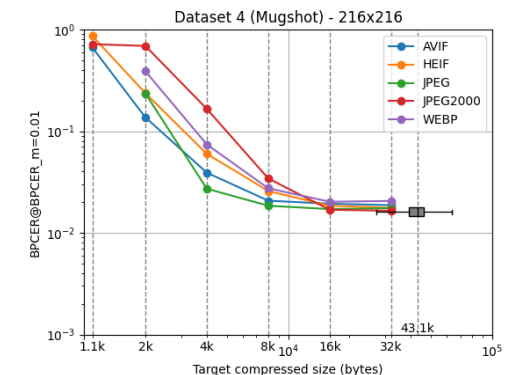
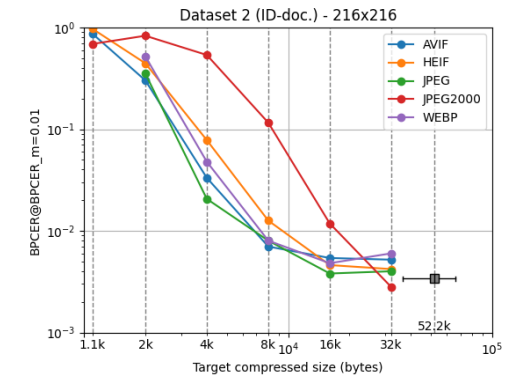
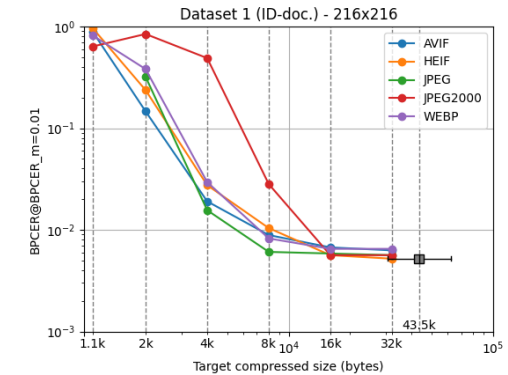
- Light compression does not increase MACER
 - an attacker can't rely on compression to mask artifacts
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How does compression affect false detections?

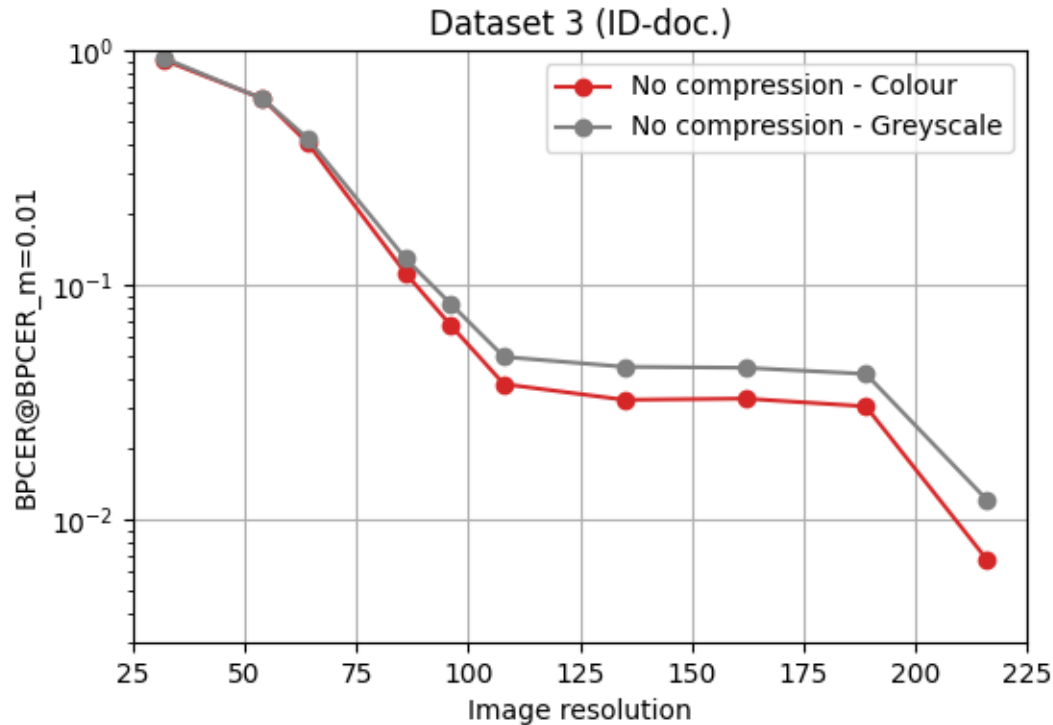


- Lower quality, compressed images cause more false positives
- We can safely compress to around 20kB

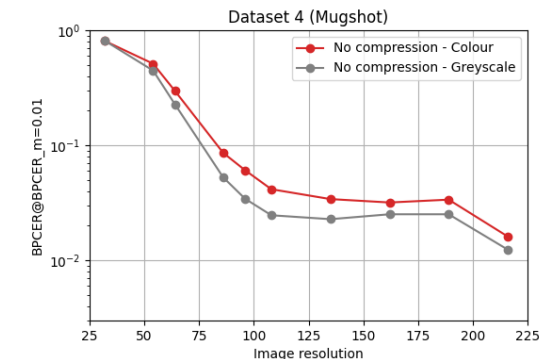
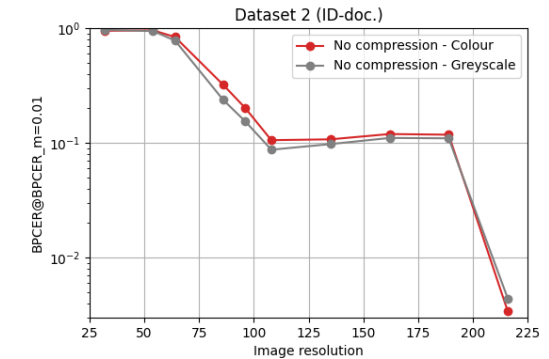
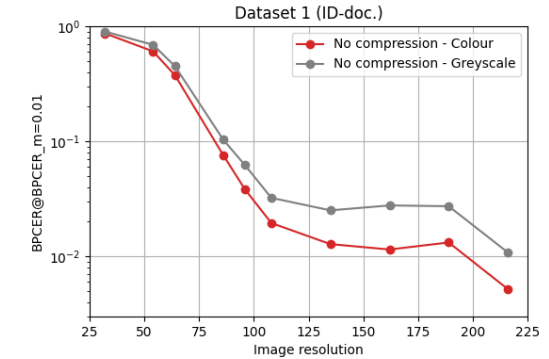


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Can we do better by pre-processing images before compression?

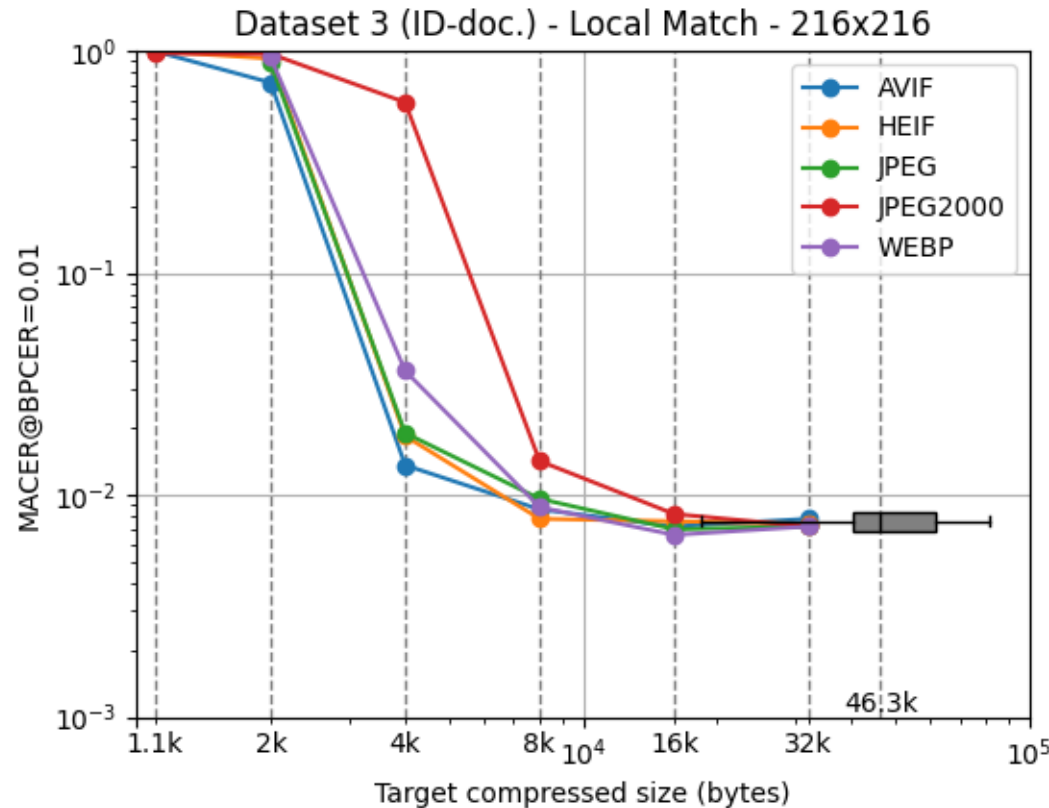


Here we see that, even with no compression, pre-processing images increases the BPCER.

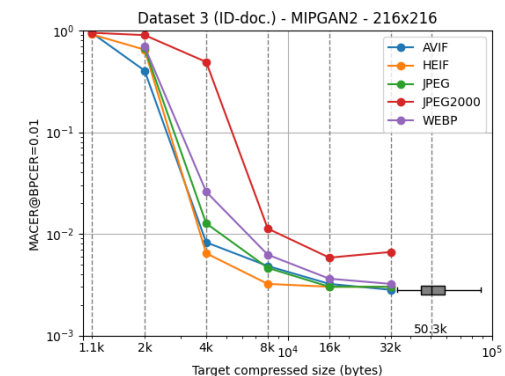
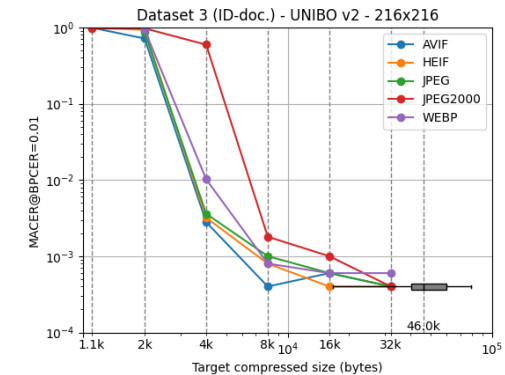
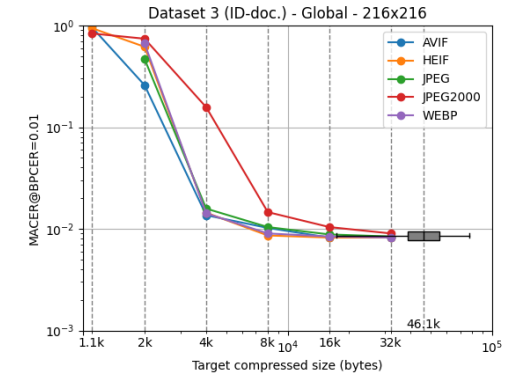


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Can we maintain performance by adapting the threshold as a function of observed file-size?



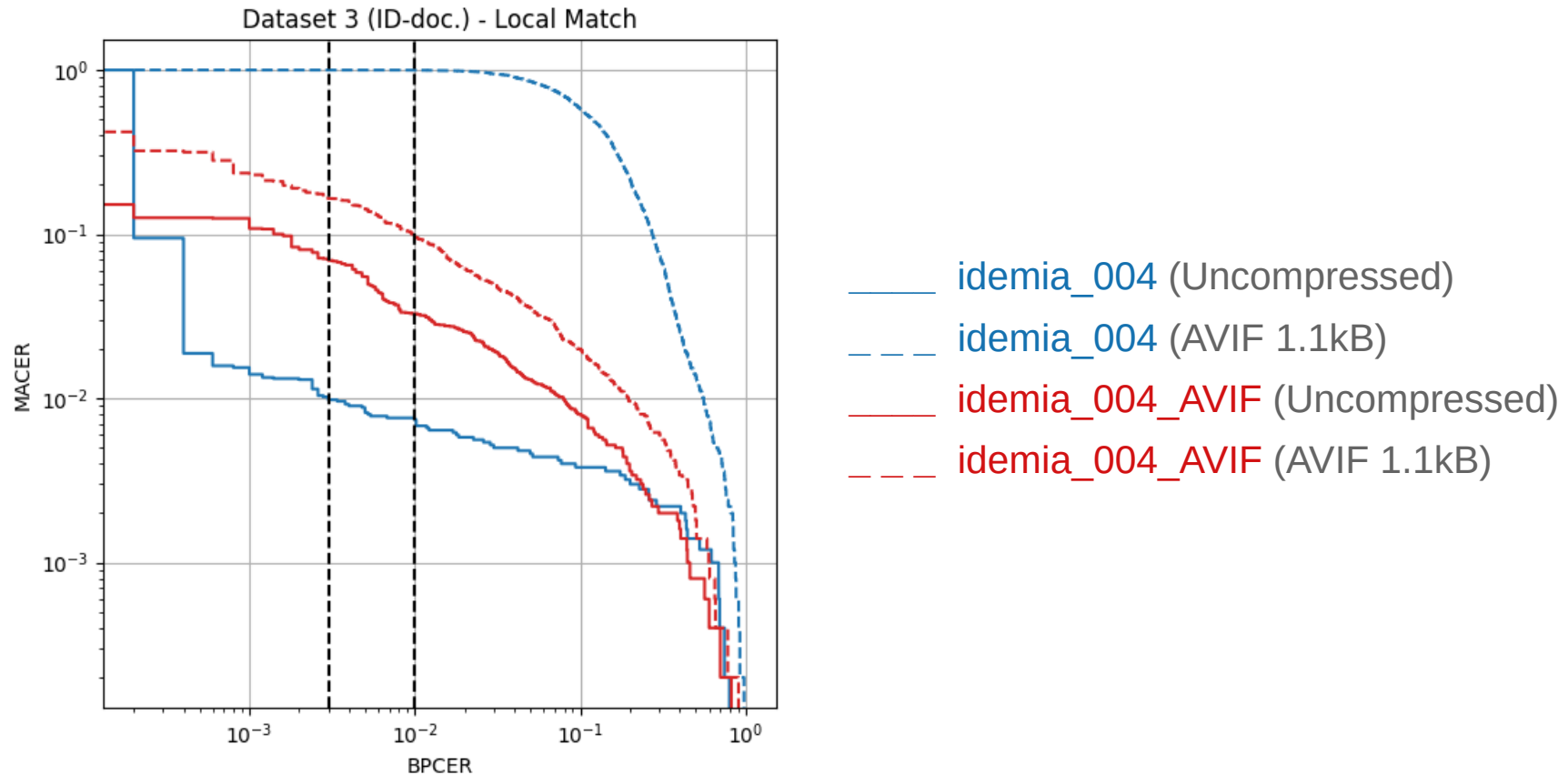
Adapting the threshold based on file-size does not help much to stabilise performance.



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Preliminary work on improving robustness to compression

We re-trained idemia_004 on images compressed to 1.1kB using AVIF.



We see that performance can be improved by re-training but that it is still far from base levels.

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Conclusions and recommendations

Conclusions:

- Compression to 1.1kB is too extreme for morph detection
 - QR codes should only be used for securely captured images
- Compression can hide morphing artifacts but also triggers increased numbers of detections (whether images are morphs or *bona fide*)
 - An attacker therefore cannot use compression to evade detection
- Performance for heavily compressed images can be improved by specialised training but is still far from base levels

Recommendations:

- Compression should not be $< \sim 20\text{kB}$ (for crop 216x216, IED=70px)
- AVIF, HEIF and WEBP formats are preferable to JPEG and JPEG2000